

Engineering Reboot 2026

Week 01 – Day 04 Field Manual

Date: June 4, 2026

Duration: 7:00 PM – 8:00 PM

Theme: Python Fundamentals

Status: Locked Baseline Version

Mission Objective

Today marks your return to software development.

Objectives:

- Verify Python installation
- Execute Python programs
- Learn variables and user input
- Build simple command-line tools
- Create notes, journal, and evidence

End State:

- Python verified
- Age Calculator completed
- Temperature Converter completed
- Evidence collected

Pre-Flight Check (7:00 PM – 7:05 PM)

Open:

- VS Code
- PowerShell

Navigate to:

C:\Projects\Engineering-Reboot-2026

Run:

python --version

Expected:

Python 3.x.x

Task 1 – Create Day 04 Files (7:05 PM – 7:10 PM)

Create:

01-Week-01\Notes\Day-04-Python-Notes.md

01-Week-01\Notes\Day-04-Journal.md

01-Week-01\Evidence\Day-04\
01-Week-01\Exercises\Day-04\

Task 2 – First Python Program (7:10 PM – 7:15 PM)

Create hello.py

Contents:
`print("Hello Glenn")`

Run:
`python hello.py`

Capture screenshot.

Task 3 – Variables and Data Types (7:15 PM – 7:25 PM)

Create variables.py

Experiment with:

- Strings
- Integers

Examples:

```
name = "Glenn"  
age = 53  
city = "Hoboken"  
years_experience = 22
```

Document observations.

Task 4 – User Input Program (7:25 PM – 7:35 PM)

Create user_input.py

Use:
`input()`
`print()`

Observe how user input changes program output.

Task 5 – Age Calculator (7:35 PM – 7:45 PM)

Create age_calculator.py

Test using:
1973

Expected:
You are 53 years old.

Task 6 – Temperature Converter (7:45 PM – 7:55 PM)

Create temperature_converter.py

Test:
0°C → 32°F
25°C → 77°F
100°C → 212°F

Notes File Requirements

Topics Covered:

- Python execution
- Variables
- Strings
- Integers
- User input
- Type conversion
- Simple calculations

Commands Used:

```
python --version
python hello.py
python variables.py
python user_input.py
python age_calculator.py
python temperature_converter.py
```

Journal File Requirements

Record:

- Accomplishments
- Challenges
- Lessons Learned
- Preparation for Day 05

Evidence Checklist

Capture screenshots showing:

- Python version
- hello.py execution
- variables.py execution
- age_calculator.py execution
- temperature_converter.py execution

End-of-Day Success Criteria

- Python verified
- hello.py executed
- variables.py executed
- user_input.py executed
- age_calculator.py completed
- temperature_converter.py completed
- Notes file completed
- Journal completed
- Evidence captured

Day 05 Preview

Project:
Command-Line Calculator

Features:

- Addition
- Subtraction
- Multiplication
- Division
- User Input
- Program Flow

This will be your first complete software artifact of the reboot.